INVERTER GEN3

HW PCBA Automated Test

Test-by-Test Analysis & Plan

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Project	Inverter Gen3 — Stage 1 HW PCBA Verification
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Firmware	HW-Test firmware (Stage 1)
Interface	PCAN-USB | CAN ID 0x100 | 250 kBit/s
Sequences	A0/A1-Sample (17 tests) | B-Sample B0/B1/B2 (21 tests)

1. Introduction

This document provides a test-by-test description of every automated test executed by the Inverter Gen3 Stage 1 HW PCBA test campaign. The purpose is to make the content of the test sequences understandable for engineers reviewing the code, planning bring-up activities, or analysing test failures.

The test campaign is implemented in Python and communicates with the board under test through a PCAN-USB adapter. All communication follows the sf_hw_test_protocol running on CAN ID 0x100 at 250 kBit/s. The board must be loaded with the HW-test firmware image (not the production firmware) before running any test.

Two test sequences are defined:

- A0/A1-Sample sequence — 17 tests — for first-article and pre-production boards with the ABI encoder interface populated.
- B-Sample sequence (B0 / B1 / B2) — 19 tests — for production boards where the ABI encoder is not fitted. Adds the pump-driver, MCU-internal, and external-temperature loopback tests that are not exercised on the A-sample.

The test outcome for each test is determined by the firmware: the host checks the V+ (verification OK) and/or S+ (self-test OK) status flags reported in the 0x81 response frame after the test timeout expires. Some tests also report ANLG[] objects whose values are compared against secondary acceptance limits.

2. Communication Protocol Overview

2.1 CAN Frame Structure

All frames use standard 11-bit CAN IDs (not extended). Every frame is exactly 8 bytes. The CAN arbitration ID is always 0x100. The first byte selects the command or response type; the remaining bytes carry one protocol object.

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Field
CMD	OBJ_INFO	VALUE[0]	VALUE[1]	VALUE[2]	VALUE[3]	FLAGS[0]	FLAGS[1]	1+1+4+2 bytes

OBJ_INFO = bits[7:5] object data type | bits[4:0] object instance (channel ID)

2.2 Request Codes (Host → Board)

Code	Name	Description
0x00	**RQST_NOP**	Do nothing (keepalive)
0x01	**RQST_SET_TEST_ENV**	Configure test environment flags (CLEAR_REBOOT=0x01, REPORT_ENABLE=0x02)
0x02	**RQST_SET_TEST**	Select and start the specified test (value = test code)
0x03	**RQST_ECHO**	Echo stimulus — firmware echoes payload back as RSPN_ECHO
0x04	**RQST_SET_PWM_FREQ**	Set PWM frequency (Hz, applies to active frequency test)
0x05	**RQST_SET_PWM_DUTY**	Set PWM duty cycle (%, applies to active duty test)
0x06	**CMD_SET_DIG_OUT**	Set digital output state
0x07	**CMD_SET_PWM_DT**	Set PWM deadtime
0x08	**CMD_SET_PWM_PULSES**	Set finite PWM pulse count
0x09	**CMD_EN_DIS_PHASE**	Enable / disable PWM branch
0x0A	**RQST_SET_PUMP_DUTY**	Set pump duty cycle
0x0B	**RQST_SET_PUMP_FREQ**	Set pump PWM frequency

2.3 Response Codes (Board → Host)

Code	Name	Description
0x80	**RSPN_NONE**	Board reports nothing (acknowledgement only)

0x81	**RSPN_TEST_STATUS**	Current test ID and status flags (primary pass/fail carrier)
0x82	**RSPN_ANALOG**	Analog measurement value (ANLG[id] with float/uint32 + flags)
0x83	**RSPN_ECHO**	Echo reply — 4-byte payload mirrors RQST_ECHO stimulus
0x84	**RSPN_BUFFER**	User buffer (4 bytes of arbitrary data)

2.4 Status Flags in RSPN_TEST_STATUS (0x81)

The FLAGS[1:0] field of a 0x81 frame carries a bitmask describing the current test state.

Bit mask	Symbol	Meaning
0x01	**BT**	Board has rebooted since last SET_TEST_ENV
0x02	**RN**	Test is currently running
0x04	**V (V- if V+ absent)**	Verification was attempted
0x08	**V+**	Verification passed ← primary PASS indicator
0x10	**S (S- if S+ absent)**	Self-test was attempted
0x20	**S+**	Self-test passed ← primary PASS indicator for self-test types

PASS rule: after the test timeout expires, the test passes if V+ (0x08) OR S+ (0x20) is set in the latest received 0x81 frame.

3. Sequence Execution Model

The automated runner replicates the state machine originally implemented in STARK_FUTURE_InvGen3_hw_test.py. The execution flow for one complete sequence is as follows:

Step	State	Action
Initialise	**State 0**	Send RQST_SET_TEST_ENV with value = REPORT_ENABLE (0x02). This tells the firmware to start reporting test status and analog objects over CAN.
Select test	**State 1**	Send RQST_SET_TEST with the next test code. Clear m_reported_objects dictionary. Record the test timeout.
Stimulus	**State 2**	If the sequence item defines a stimulus_type (only HW_TEST_CAN_ECHO does this), send that stimulus frame immediately after SET_TEST.
Wait	**State 3**	Decrement the timeout counter by the sequence quantum (20 ms per loop tick). Accumulate all received 0x82 / 0x81 / 0x83 / 0x84 frames in m_reported_objects. When timeout reaches zero, advance.
Evaluate	**Post-wait**	Read the last 0x81 frame from m_reported_objects. Check V+ (bit 3) or S+ (bit 5) in FLAGS. Check analog limits on all received ANLG objects. Record PASS or FAIL.
Next test	**—**	Return to State 1 with the next item in the sequence list. Repeat until all tests are done.
Finish	**State 4**	Set m_sequenceInProgress = False. Write results CSV. Generate Word report.

Quantum timing: the sequence thread sleeps for 20 ms between state transitions (APP_HW_TEST_SEQ_QUANTUM_TIME_IN_MS = 20). All timeout values in the test list are in milliseconds; they are decremented by 20 on each pass.

Per-test execution time equals the configured Timeout: the runner always decrements through the full quantum countdown before evaluating results (no early exit). Realistic wall-clock per campaign also includes a one-shot boot wait (1.0 s) and env-setup wait (0.3 s), plus a 0.2 s inter-test settling gap between consecutive tests.
Early-exit (config.EARLY_EXIT_ON_PASS = True, default): once the latest 0x81 frame reports V+ (verification OK) or S+ (self-test OK), the runner drops out of the countdown after a short grace window (EARLY_EXIT_GRACE_MS = 200 ms) — keeping any trailing analog frames — instead of waiting the full timeout. Monitoring-only tests (no V/S bits) still wait the full timeout. This typically halves campaign duration.

Firmware OUT_OF_RANGE flag — every analog frame (0x82) carries a 2-byte flags field; bit 0 is the firmware's own OUT_OF_RANGE indicator, set whenever a measurement falls outside the firmware-defined valid range for that channel. The runner now surfaces this verdict with a '*' marker in the per-test analog listing. A '*' is independent of the Python limit-check (shown with '!!'): the firmware's OUT_OF_RANGE flag is what determines the firmware's V+/V- verdict, while '!!' indicates a value outside the secondary limits in hw_test_criteria.py. Both can appear on the same line.

4. Test Sequences

4.1 A0/A1-Sample Sequence (17 tests)

Total campaign duration (sum of per-test execution times): 27000 ms (≈ 27.0 s). Realistic wall-clock including 1.0 s boot wait + 0.3 s env setup + 17 × 0.2 s inter-test gaps: ≈ 31.7 s.

#	Test Name	Code	Timeout	Stim.	Pass Criterion
1	LEDS	0x01	500 ms	—	V+ flag set (firmware confirms LED cycle completed without fault). No ANLG[] obj…
2	SUPPLY_VOLTAGES	0x02	500 ms	—	V+ flag set (all measured rails within firmware acceptance window). ANLG[15] = 5…
3	ALL_ADC_MEASUREMENTS	0x03	500 ms	—	V+ flag set. Key channel limits: ANLG[1-3] (I_Ph U/V/W) within [-5 A, +5 A] at i…
4	PWM_DRIVERS_USER_MEASUREMENTS	0x05	500 ms	—	V+ flag set (isolated ADC reads within expected range per firmware). ANLG[] valu…
5	MEASUREMENTS_DIAGNOSTICS	0x08	1000 ms	—	V+ flag set (all diagnostic checks passed). Longer timeout (1000 ms) because dia…
6	ALL_PWM_SUPPLIES	0x0C	1500 ms	—	V+ flag set (all three driver supply rails confirmed present). Extra 500 ms comp…
7	ALL_DRIVERS_STATUS	0x10	3500 ms	—	V+ flag set (all drivers report no fault condition, status bits match expected v…
8	ALL_DRIVERS_SETTING	0x14	3500 ms	—	V+ flag set (all SPI write/read-back comparisons pass). Long timeout to allow mu…
9	ALL_DRIVERS_FREQUENCY	0x18	4000 ms	—	V+ flag set (frequency configuration accepted and verified). 4000 ms timeout all…
10	ALL_DRIVERS_DUTY	0x19	4000 ms	—	V+ flag set (duty-cycle programming verified for all phases). Long timeout (4000…
11	CAN_ECHO	0x1E	1000 ms	ECHO	RSPN_ECHO (0x83) received with identical 4-byte payload to stimulus. Additional:…

12	UART_CHECK_LOOPBACK	0x21	1000 ms	—	S+ flag set (UART self-test passed — transmitted pattern matches received echo).…
13	I2C_TEMP_HUMID_AUTO_TX	0x22	1000 ms	—	V+ flag set (sensor acknowledged on I2C bus, valid reading obtained). Sensor val…
14	SPI_FLASH_MEM_AUTO_TX	0x23	1000 ms	—	V+ flag set (JEDEC ID recognised, write/read-back match). A missing or wrong-typ…
15	ENC_SPI_AUTO_TX	0x24	500 ms	—	V+ flag set (SPI encoder responds with valid framing and within expected data ra…
16	ENC_ABI_LOOPBACK	0x27	2500 ms	—	S+ flag set (self-test: generated vs received ABI pulse count within tolerance).…
17	EXTRA_GPIOs_LOOPBACK	0x29	500 ms	—	V+ flag set (all GPIO loopback reads match driven values). A missing loopback co…
	TOTAL — execution time (sum of per-test timeouts)		**27000 ms (≈ 27.0 s)**		**Add ≈ 1.3 s boot/env setup + 0.2 s × N inter-test gaps for total wall-clock**

4.2 B-Sample Sequence — B0 / B1 / B2 (21 tests)

Identical to the A0/A1 sequence minus HW_TEST_ENC_ABI_LOOPBACK and HW_TEST_ENC_SPI_AUTO_TX (B-sample boards use the sin/cos encoder front-end only), plus five B-sample additions: HW_TEST_MCU_INTER_MEASUREMENTS (0x04), the sin/cos analog loopbacks (0x25, 0x26), the unipolar and bipolar voltage conditioning loopbacks (0x2A, 0x2B), and HW_TEST_PUMP_AUTO (0x2F). HW_TEST_EXT_TEMPERATURE_LOOPBACK (0x2E) is intentionally NOT in the default sequence because only EXT_TEMP1/EXT_TEMP2 are populated on B0/B1; the firmware's DAC sweep across all four NTC paths would always saturate the unconnected ones. B0, B1 and B2 share the same sequence. Worst-case per-test execution time: 36 500 ms (≈ 36.5 s). Realistic wall-clock including 1.0 s boot wait + 0.3 s env setup + 21 × 0.2 s inter-test gaps: ≈ 42.0 s — typically ~50 % less when early-exit on a passing last-N window is enabled (default).

#	Test Name	Code	Timeout	Stim.	Pass Criterion
1	LEDS	0x01	500 ms	—	V+ flag set (LED cycle completed without fault); visual confirmation also recommended.
2	SUPPLY_VOLTAGES	0x02	500 ms	—	V+ flag set (all rails within firmware acceptance window). ANLG[15]=5V_RAIL in [4.75, 5.25] V; ANLG[14]=SUPPLY_28V in [25.0, 31.0] V.
3	ALL_ADC_MEASUREMENTS	0x03	500 ms	—	V+ flag set. ANLG[1-3] I_Ph U/V/W in [-5, +5] A at idle; ANLG[4] DC_LINK in

					expected range; ANLG[13]=PCB_TEMP.
4	MCU_INTER_MEASUREMENTS	0x04	500 ms	—	V+ flag set. ANLG[18]=VDDCORE in [1.30, 1.40] V; ANLG[20]=VREFINT in [3.00, 3.60] V; ANLG[21]=VBAT in [2.80, 3.60] V; ANLG[19]=MCU_TEMP in [-10, 85] °C.
5	PWM_DRIVERS_USER_MEASUREMENTS	0x05	500 ms	—	V+ flag set. ANLG[22-23]=PCB_TEMP0/1, ANLG[24-26]=PM_U/V/W_TEMP, all in [-10, 85] °C.
6	MEASUREMENTS_DIAGNOSTICS	0x08	1000 ms	—	V+ flag set (all diagnostic checks passed).
7	ALL_PWM_SUPPLIES	0x0C	1500 ms	—	V+ flag set (all three driver supply rails confirmed present).
8	ALL_DRIVERS_STATUS	0x10	3500 ms	—	V+ flag set (all drivers report no fault condition).
9	ALL_DRIVERS_SETTING	0x14	3500 ms	—	V+ flag set (all SPI write/read-back comparisons pass).
10	ALL_DRIVERS_FREQUENCY	0x18	4000 ms	—	V+ flag set (frequency configuration accepted and verified).
11	ALL_DRIVERS_DUTY	0x19	4000 ms	—	V+ flag set (duty-cycle programming verified for all phases).
12	CAN_ECHO	0x1E	1000 ms	ECHO	RSPN_ECHO (0x83) received; payload matches stimulus.
13	UART_CHECK_LOOPBACK	0x21	1000 ms	—	S+ flag set (UART self-test passed).
14	I2C_TEMP_HUMID_AUTO_TX	0x22	1000 ms	—	V+ flag set (I2C sensor acknowledged, plausible readings).
15	SPI_FLASH_MEM_AUTO_TX	0x23	1000 ms	—	V+ flag set (JEDEC ID recognised, write/read-back match).
16	ENC_SINCOS_SIN_LOOPBACK	0x25	1500 ms	—	V+ flag set (DAC→ADC tracking for the SIN differential analog front-end). Common-mode + range

					sweep covered. ANLG[16]=ENC_SIN.
17	ENC_SINCOS_COS_LOOPBACK	0x26	1500 ms	—	V+ flag set (DAC→ADC tracking for the COS differential analog front-end). Same coverage as 0x25 on the COS leg. ANLG[17]=ENC_COS.
18	EXTRA_GPIOs_LOOPBACK	0x29	500 ms	—	V+ flag set (all GPIO loopback reads match driven values).
19	POWER_UNIPLR_V_LOOPBACK	0x2A	2000 ms	—	V+ flag set (unipolar voltage conditioning chain verified through isolated-input loopback). Targets DC-link voltage sense chain.
20	POWER_BIPLR_V_LOOPBACK	0x2B	2000 ms	—	V+ flag set (bipolar voltage conditioning chain — phase-to-DCneg voltage sense UV/WV — verified through isolated-input loopback).
21	PUMP_AUTO	0x2F	5000 ms	—	V+ flag set (firmware sweeps EN/LATCH/SEL combinations + duty cycle; pump HS-switches report nominal). ANLG[32]/ANLG[34] pump HS temperatures, ANLG[39] PUMP_SPEED_TACH.
	TOTAL — execution time (worst-case sum of per-test timeouts)		**36500 ms (≈ 36.5 s)**		**Early-exit on a passing last-N window typically halves this; add ≈ 1.3 s boot/env setup + 0.2 s × N inter-test gaps for total wall-clock.**

4.3 3-Phase Campaign Structure (B-sample)

Operationally the 21-test B-sample sequence is split into THREE independently runnable phases by how each test needs to be driven: Phase 1 (no operator input), Phase 2 (external scope verification, probe moves) and Phase 3 (operator DAC hookups). The verify_pcba.py orchestrator runs each phase under its own flag and merges the results into one unified ValidationReport workbook.

Phase 1 — Self auto-verification (16 tests, sequence "B<x>_SELF"). Runs unattended over CAN once the board is powered and the standard test-fixture jumpers (UART TX-RX, GPIO loopback) are in place: SUPPLY_VOLTAGES, ALL_ADC_MEASUREMENTS, MCU_INTER_MEASUREMENTS, PWM_DRIVERS_USER_MEASUREMENTS, MEASUREMENTS_DIAGNOSTICS, ALL_PWM_SUPPLIES, ALL_DRIVERS_STATUS, ALL_DRIVERS_SETTING, ALL_DRIVERS_FREQUENCY (0x18), ALL_DRIVERS_DUTY (0x19), CAN_ECHO, UART_CHECK_LOOPBACK, I2C_TEMP_HUMID_AUTO_TX, SPI_FLASH_MEM_AUTO_TX, EXTRA_GPIOs_LOOPBACK, PUMP_AUTO. CLI: --self (default on) / --no-self to skip. Note: PUMP_AUTO will report FAIL on a bench where no pump is wired to the pump output — this is expected and the result is recorded as-is in the report.

Phase 2 — Power modules (1 firmware test, HW_TEST_ALL_POWER_MODULE 0x1B; scope-assisted, see §6.5). Sequence is just that one test, but its execution is a 6 switches × 9 setpoint sweep (10/20/30 kHz × 25/50/75 % at 1 µs dead-band). After each probe move the test is cycled (SET_TEST(HW_TEST_NO_TEST) → SET_TEST(HW_TEST_ALL_POWER_MODULE) + re-pin the dead-band) so the gate drivers are reset and the "not ready" condition caused by the physical probe change is cleared before the next sweep. No initial warm-up is required (the per-switch reset replaces it). CLI: --power-module.

Phase 3 — Operator-verified tests (5 tests, sequence "B<x>_LOOPBACK"). Each test requires the operator to do something physical before/during the test runs. HW_TEST_LEDS (0x01) — operator visually verifies LED1/LED2 blink during the test (a pre-test prompt instructs the operator and waits for ENTER). The four DAC-injection loopbacks — ENC_SINCOS_SIN_LOOPBACK (0x25), ENC_SINCOS_COS_LOOPBACK (0x26), POWER_UNIPLR_V_LOOPBACK (0x2A), POWER_BIPLR_V_LOOPBACK (0x2B) — each require the operator to wire a DAC output to an analog input before the test runs. The runner prints the hookup/observation instruction and waits for ENTER per test, always interactive (the --no-prompts flag does not suppress these prompts). CLI: --loopback. Implementation note: the protocol method is sfHwTestProtocolGetBSampleOperatorVerifiedTestsList (older callers can still use the alias sfHwTestProtocolGetBSampleLoopbackTestsList).

Phase 1 ∪ Phase 3 is exactly the full B-sample list (the two are disjoint and together exhaustive); Phase 2 sits alongside and is not in either base list. The --all flag enables all three phases in a single verify_pcba.py invocation. The unified ValidationReport_<SN>_<ts>.xlsx tags every test row with its Phase number, in addition to the dedicated Power Module Sweep sheet for Phase 2.

5. Test-by-Test Analysis

Each subsection below covers one unique test in detail. Tests are ordered by CAN test code.

5.1 HW_TEST_LEDS (0x01)

Test	Code 0x01 | Timeout 500 ms | Sequences: Both
Description	LED continuous test. Firmware drives all indicator LEDs in a repeating pattern so the operator can visually verify all LEDs are installed and functional.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (firmware confirms LED cycle completed without fault). No ANLG[] objects reported.

ANLG[] Channels	None
Notes	Run last in both sequences. Visual confirmation also recommended while timeout is active.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x01. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.2 HW_TEST_SUPPLY_VOLTAGES (0x02)

Test	Code 0x02 | Timeout 500 ms | Sequences: Both
Description	Supply voltage rail check. Firmware measures all regulated DC rails (3.3 V, 5 V, 12 V, etc.) through internal ADC channels and reports them as ANLG[] objects. Rails outside tolerance are flagged.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all measured rails within firmware acceptance window). ANLG[15] = 5 V rail expected within [4.75 V, 5.25 V].
ANLG[] Channels	ANLG[15] = 5V_RAIL
Notes	First functional test in both sequences. ANLG out-of-range flag (bit 0 of analog flags) is set by firmware if any rail is outside limit.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x02. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.3 HW_TEST_ALL_ADC_MEASUREMENTS (0x03)

Test	**Code 0x03 | Timeout 500 ms | Sequences: Both**
Description	Full ADC scan. Firmware samples every main-board ADC channel — phase currents, DC-link voltage, PCB temperatures — and reports them all as a stream of ANLG[] objects over CAN.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set. Key channel limits: ANLG[1-3] (I_Ph U/V/W) within [-5 A, +5 A] at idle; ANLG[4] (DC_LINK) within expected range; ANLG[13]=PCB_TEMP within [-10 °C, 85 °C]; ANLG[14]=SUPPLY_28V within [25.0, 31.0] V.
ANLG[] Channels	ANLG[0]=HW_VERSION, ANLG[1-3]=I_Ph_U/V/W, ANLG[4]=DC_LINK, ANLG[5-6]=UV/WV_VOLTAGE, ANLG[13]=PCB_TEMP, ANLG[14]=SUPPLY_28V, ANLG[15]=SUPPLY_5V, ANLG[16-17]=ENC_SIN/COS, ANLG[18-21]=MCU rails, ANLG[32, 34]=PUMP_HS1/2_TEMP (remapped from PUMP_SNS1/2), ANLG[35-38]=EXT_TEMP1-4 (remapped from internal IDs 7-10).
Notes	Rich ANLG[] stream — expect multiple 0x82 frames before 0x81 with V+. Firmware remaps PUMP_SNS1/2 (11,12) → 32/34 and EXT_TEMP1-4 (7-10) → 35-38 on the wire; Python channel limits are placed at the reported IDs accordingly. PCB_TEMP0/1 (ANLG[22, 23]) are reported by test 0x05, NOT by 0x03. Current failure mode on B1: firmware ranges HW_VERSION (channel 0) and the four EXT_TEMP channels; with only EXT_TEMP1/EXT_TEMP2 populated on B1 the firmware always flags EXT_TEMP3 and EXT_TEMP4 as OUT_OF_RANGE ('*' marker). HW_VERSION is also reported as OUT_OF_RANGE, which is a firmware oversight (the version index is a valid identifier, not a ranged measurement). These three '*' flags drive the firmware verdict to V- and the test to FAIL even though every other channel passes. Python's hw_test_criteria.SKIP_CHANNELS_BY_HW_VERSION ignores the two unpopulated EXT_TEMPs at the limit-check level but cannot override the firmware V-; the FW side will need its own HW-version aware skip in analog_measurements_test() to pass cleanly on B1.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x03. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.

4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.4 HW_TEST_MCU_INTER_MEASUREMENTS (0x04)

Test	**Code 0x04 | Timeout 500 ms | Sequences: B-Sample (B0/B1/B2)**
Description	MCU-internal measurement scan. Firmware samples the on-chip ADC channels connected to VDDCORE, VREFINT, VBAT and the internal temperature sensor and reports each as an ANLG[] object.
Sequence(s)	B-Sample (B0/B1/B2) only
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all four MCU internal measurements within firmware acceptance window). Python secondary limits: VDDCORE [1.30, 1.40] V, VREFINT [3.00, 3.60] V, VBAT [2.80, 3.60] V, MCU_TEMP [-10, 85] °C.
ANLG[] Channels	ANLG[18]=VDD_CORE, ANLG[19]=MCU_TEMP, ANLG[20]=VREF_INT, ANLG[21]=VBAT_INT.
Notes	Only test in the campaign that emits the MCU-internal channels. A drifting VDDCORE or VBAT is a strong indicator of MCU power-supply decoupling problems or a bad PCB build.

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x04. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Four analog reports (VDDCORE, MCU_TEMP, VREFINT, VBAT) round-robin during the timeout window.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.5 HW_TEST_PWM_DRIVERS_USER_MEASUREMENTS (0x05)

Test	**Code 0x05 | Timeout 500 ms | Sequences: Both**
Description	Isolated ADC measurements. Each gate-driver board carries an isolated ADC; this test polls those ADCs and reports their readings. Verifies the driver auxiliary power supply and isolated measurement chain.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (isolated ADC reads within expected range per firmware). ANLG[] values from driver-side ADCs expected.
ANLG[] Channels	ANLG[22]=PCB_TEMP0, ANLG[23]=PCB_TEMP1, ANLG[24]=POWER_MODULE_U_TEMP, ANLG[25]=POWER_MODULE_V_TEMP, ANLG[26]=POWER_MODULE_W_TEMP.
Notes	All five reported channels expected within [-10 °C, 85 °C]. ANLG[27]=PMW_DCLINK is obsolete (kept for legacy logs only).

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x05. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.6 HW_TEST_MEASUREMENTS_DIAGNOSTICS (0x08)

Test	**Code 0x08 | Timeout 1000 ms | Sequences: Both**
Description	Measurement signal integrity diagnostics. Firmware exercises diagnostic routines on the ADC signal chain — checks for short circuits, open inputs, noise levels, and sensor health. Reports diagnostic status objects.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all diagnostic checks passed). Longer timeout (1000 ms) because diagnostics include settling time.
ANLG[] Channels	Diagnostic status objects; may include analog readings
Notes	If any sensor is disconnected or shorted, V+ will not be set.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x08. m_reported_objects cleared. Test timeout = 1000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.7 HW_TEST_ALL_PWM_SUPPLIES (0x0C)

Test	**Code 0x0C | Timeout 1500 ms | Sequences: Both**
Description	Gate-driver power-supply enable test (all three phases simultaneously). Firmware enables all three PWM power stages, verifies the enable logic, and confirms supply voltage appears on driver rails.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all three driver supply rails confirmed present). Extra 500 ms compared to simpler tests to allow supply stabilisation.
ANLG[] Channels	Driver supply rail ANLG[] channels if reported
Notes	Tests PWM1, PWM2, and PWM3 supplies together (code 0x0C = all three combined). Individual supply tests 0x09–0x0B are not part of the automated sequence.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x0C. m_reported_objects cleared. Test timeout = 1500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.8 HW_TEST_ALL_DRIVERS_STATUS (0x10)

Test	**Code 0x10 | Timeout 3500 ms | Sequences: Both**
Description	Gate-driver fault-status check (all phases). Firmware reads the nFAULT and status registers of all three gate drivers for multiple operating conditions (power-on, enabled, after fault injection) and reports pass/fail per driver.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all drivers report no fault condition, status bits match expected values). Long timeout (3500 ms) because firmware exercises several fault/clear cycles per driver.
ANLG[] Channels	Possibly driver status ANLG objects
Notes	Individual driver tests 0x0D–0x0F are not in the sequence; 0x10 covers all three. A stuck nFAULT line will fail this test.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x10. m_reported_objects cleared. Test timeout = 3500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.9 HW_TEST_ALL_DRIVERS_SETTING (0x14)

Test	Code 0x14 | Timeout 3500 ms | Sequences: Both
Description	Gate-driver configuration register write/read-back verification. Firmware writes known configuration patterns to each driver's SPI registers and reads them back, confirming the SPI link and driver programmability.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all SPI write/read-back comparisons pass). Long timeout to allow multiple register access cycles per driver.
ANLG[] Channels	None expected
Notes	SPI communication integrity is critical here. A wrong soldering on SPI lines will produce a V- result.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x14. m_reported_objects cleared. Test timeout = 3500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.10 HW_TEST_ALL_DRIVERS_FREQUENCY (0x18)

Test	**Code 0x18 | Timeout 4000 ms | Sequences: Both**
Description	PWM frequency configuration test (all phases). Firmware programs the PWM timers to a reference frequency, verifies the timer reload register, and confirms the PWM output frequency through the driver feedback path.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (frequency configuration accepted and verified). 4000 ms timeout allows firmware to step through multiple frequency points.
ANLG[] Channels	Timing/frequency ANLG objects if reported
Notes	Performed before HW_TEST_ALL_DRIVERS_DUTY. Firmware keeps switching inactive (no high-voltage needed).

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x18. m_reported_objects cleared. Test timeout = 4000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.11 HW_TEST_ALL_DRIVERS_DUTY (0x19)

Test	**Code 0x19 | Timeout 4000 ms | Sequences: Both**
Description	PWM duty-cycle configuration test (all phases). Firmware programs a range of duty-cycle values to the PWM compare registers, confirms register acceptance, and steps through several duty points.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (duty-cycle programming verified for all phases). Long timeout (4000 ms) for multi-step duty sweep.
ANLG[] Channels	None expected
Notes	No switching occurs during this test; it only verifies register-level configuration.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x19. m_reported_objects cleared. Test timeout = 4000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.12 HW_TEST_CAN_ECHO (0x1E)

Test	Code 0x1E | Timeout 1000 ms | Sequences: Both
Description	CAN echo test. Firmware waits for an ECHO request frame (RQST_ECHO, code 0x03) from the host. Upon reception it reflects the 4-byte payload unchanged back as a RSPN_ECHO (code 0x83) frame. The host confirms its own stimulus was echoed correctly.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	Type: SF_HW_TEST_PROTOCOL_RQST_ECHO | Value: 0x55 (pattern repeated in bytes 2-5)
Pass Criteria	RSPN_ECHO (0x83) received with identical 4-byte payload to stimulus. Additional: V+ flag in 0x81 response confirms firmware side is satisfied. Buffer object in m_reported_objects will contain echoed bytes.
ANLG[] Channels	BUFFER object (0x84 frame) with echoed payload
Notes	Only test in both sequences that sends an explicit stimulus after SET_TEST. State machine: State 0→SET_TEST_ENV, State 1→SET_TEST(0x1E)+clear objects, State 2→send RQST_ECHO(0x55×4), State 3→wait 1000 ms.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x1E. m_reported_objects cleared. Test timeout = 1000 ms.
3 →	**SF_HW_TEST_PROTOCOL_RQST_ECHO (0x03)**	Stimulus value = 0x55 (pattern repeated in bytes 2-5). Sent immediately after SET_TEST.
4 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
5 ←	**RSPN_ECHO (0x83)**	Echo reply — 4-byte payload mirrors the RQST_ECHO stimulus.
6 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.13 HW_TEST_UART_CHECK_LOOPBACK (0x21)

Test	**Code 0x21 | Timeout 1000 ms | Sequences: Both**
Description	UART loopback self-test. Firmware transmits a known byte sequence on the UART Tx pin and expects to receive an exact echo on the Rx pin (hardware loopback jumper or internal loopback must be in place).
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	S+ flag set (UART self-test passed — transmitted pattern matches received echo). V+ may also be set.
ANLG[] Channels	None
Notes	Requires a physical loopback jumper between UART Tx and Rx (or internal MCU loopback if supported). A missing jumper will produce an S- result.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x21. m_reported_objects cleared. Test timeout = 1000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.14 HW_TEST_I2C_TEMP_HUMID_AUTO_TX (0x22)

Test	**Code 0x22 | Timeout 1000 ms | Sequences: Both**
Description	I2C temperature/humidity sensor polling. Firmware communicates with the external I2C sensor (e.g., SHT-series), reads temperature and humidity, and reports the values as ANLG[] objects continuously.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (sensor acknowledged on I2C bus, valid reading obtained). Sensor values within physically plausible range.
ANLG[] Channels	ANLG objects for ambient temperature and humidity (channel IDs TBD by firmware)
Notes	If the I2C sensor is not populated or there is a bus fault, firmware will not receive an ACK and V+ will not be set.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x22. m_reported_objects cleared. Test timeout = 1000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.15 HW_TEST_SPI_FLASH_MEM_AUTO_TX (0x23)

Test	**Code 0x23 | Timeout 1000 ms | Sequences: Both**
Description	External SPI flash memory access test. Firmware performs representative transactions on the external SPI NOR flash: read JEDEC ID, write a test page, read it back, verify data integrity.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (JEDEC ID recognised, write/read-back match). A missing or wrong-type flash device will fail this test.
ANLG[] Channels	None expected
Notes	Flash transactions involve chip-erase or page-write — can take up to 1000 ms depending on device timing.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x23. m_reported_objects cleared. Test timeout = 1000 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.16 HW_TEST_ENC_SPI_AUTO_TX (0x24)

Test	**Code 0x24 | Timeout 500 ms | Sequences: A0/A1 only**
Description	Encoder SPI interface transactions. Firmware communicates with the SPI-based encoder interface, reads the encoder register set, and verifies the response is within expected protocol framing.
Sequence(s)	A0/A1-Sample only — B-Sample uses the sin/cos encoder front-end (0x25/0x26) and does not exercise the SPI encoder.
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (SPI encoder responds with valid framing and within expected data range). A disconnected or faulty encoder SPI will result in V-.
ANLG[] Channels	Possibly encoder position/status ANLG objects
Notes	Distinct from the ABI loopback test. SPI encoder uses separate lines from the ABI differential inputs.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x24. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.17 HW_TEST_ENC_SINCOS_SIN_LOOPBACK (0x25)

Test	Code 0x25 | Timeout 1500 ms | Sequences: B-Sample (B0/B1/B2) only
Description	Encoder SIN differential analog front-end loopback. Firmware drives the internal DAC across the full input range of the SIN+/SIN− differential pair and verifies the corresponding ADC reads track the stimulus through positive, negative and common-mode operating points.
Sequence(s)	B-Sample (B0/B1/B2) only
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (SIN front-end DAC→ADC tracking within tolerance over the swept range). ANLG[16]=ENC_SIN instantaneous value reported.
ANLG[] Channels	ANLG[16] = ENC_SIN
Notes	Counterpart to 0x26 (COS leg). A V- result points to a broken sin-cos op-amp, biasing resistor or ADC mux on the sin-cos analog stage (sheet EncoderSinCosAnalogStage).

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x25. m_reported_objects cleared. Test timeout = 1500 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Analog reports during the DAC sweep / test execution.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion (or early-exit on V+/S+) — FLAGS field contains V+/S+ bits.

5.18 HW_TEST_ENC_SINCOS_COS_LOOPBACK (0x26)

Test	Code 0x26 | Timeout 1500 ms | Sequences: B-Sample (B0/B1/B2) only
Description	Encoder COS differential analog front-end loopback. Same procedure as 0x25 applied to the COS+/COS− leg.
Sequence(s)	B-Sample (B0/B1/B2) only
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (COS front-end DAC→ADC tracking within tolerance). ANLG[17]=ENC_COS instantaneous value reported.
ANLG[] Channels	ANLG[17] = ENC_COS
Notes	Run immediately after 0x25 so any wiring or analog-stage failure is localised between the two legs.

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x26. m_reported_objects cleared. Test timeout = 1500 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Analog reports during the DAC sweep / test execution.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion (or early-exit on V+/S+) — FLAGS field contains V+/S+ bits.

5.19 HW_TEST_ENC_ABI_LOOPBACK (0x27)

Test	Code 0x27 | Timeout 2500 ms | Sequences: A0/A1

Description	ABI (incremental) encoder loopback test. Firmware generates a known ABI pulse train on internal test outputs and routes it through the ABI differential input front-end. It then counts the received pulses and verifies the count matches the generated pattern.
Sequence(s)	A0/A1-Sample ONLY
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	S+ flag set (self-test: generated vs received ABI pulse count within tolerance). Long timeout (2500 ms) for multiple pulse burst verifications.
ANLG[] Channels	Encoder position/count ANLG objects
Notes	A0/A1-Sample ONLY — not included in B0 sequence. Longer timeout because firmware needs to generate and verify multiple pulse trains at different frequencies.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x27. m_reported_objects cleared. Test timeout = 2500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.20 HW_TEST_EXTRA_GPIOs_LOOPBACK (0x29)

Test	**Code 0x29 | Timeout 500 ms | Sequences: Both**
Description	Extra GPIO loopback test. Firmware sets each extra general-purpose I/O output high then low and reads back through a corresponding input (hardware loopback required) to verify digital I/O path continuity.
Sequence(s)	A0/A1-Sample and B-Sample (B0/B1/B2)
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (all GPIO loopback reads match driven values). A missing loopback connection or a stuck GPIO will result in V-.
ANLG[] Channels	None
Notes	Requires a loopback jumper connecting the extra GPIO output(s) to input(s) as defined in the hardware test fixture specification.

CAN Frame Exchange

Step	Frame	Content
1 →	**RQST_SET_TEST_ENV (0x01)**	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	**RQST_SET_TEST (0x02)**	value = 0x29. m_reported_objects cleared. Test timeout = 500 ms.
3 ←	**RSPN_ANALOG (0x82) ×N**	Zero or more analog measurement frames during the timeout window. Values accumulated in m_reported_objects.
4 ←	**RSPN_TEST_STATUS (0x81)**	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

5.21 HW_TEST_POWER_UNIPLR_V_LOOPBACK (0x2A)

Test	Code 0x2A | Timeout 2000 ms | Sequences: B-Sample (B0/B1/B2) only
Description	Unipolar voltage-conditioning loopback (DC-link voltage sense chain). Firmware drives the DAC into the isolated input of the unipolar voltage conditioning op-amp chain and verifies the resulting ADC reads cover the full expected range without saturation or non-linearity.
Sequence(s)	B-Sample (B0/B1/B2) only
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (unipolar chain DAC→ADC tracking within tolerance). Covers the DC-link voltage path (VoltageConditioningUnipolar sheet).
ANLG[] Channels	ANLG[4] = DC_LINK (during stimulus sweep)
Notes	Verifies signal integrity from the isolated voltage front-end through the conditioning chain to the MCU ADC, without requiring high-voltage DC bus.

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x2A. m_reported_objects cleared. Test timeout = 2000 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Analog reports during the DAC sweep / test execution.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion (or early-exit on V+/S+) — FLAGS field contains V+/S+ bits.

5.22 HW_TEST_POWER_BIPLR_V_LOOPBACK (0x2B)

Test	Code 0x2B | Timeout 2000 ms | Sequences: B-Sample (B0/B1/B2) only
Description	Bipolar voltage-conditioning loopback (phase-to-DCneg voltage sense chain — UV and WV voltages). Firmware drives the DAC across both polarities into the isolated input and verifies the ADC reads cover positive, negative and zero crossings without rail-clipping.
Sequence(s)	B-Sample (B0/B1/B2) only
Stimulus	None — no additional CAN frame is sent after SET_TEST
Pass Criteria	V+ flag set (bipolar chain DAC→ADC tracking within tolerance, both polarities). Covers UV/WV phase-to-DCneg voltage paths (VoltageConditioningPhaseToDCNeg sheet).
ANLG[] Channels	ANLG[5] = UV_VOLTAGE, ANLG[6] = WV_VOLTAGE
Notes	Complements 0x2A: where 0x2A covers the unipolar DC rail sense, 0x2B exercises the bipolar phase-to-DC- neg sense chain used for motor terminal voltages.

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x2B. m_reported_objects cleared. Test timeout = 2000 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Analog reports during the DAC sweep / test execution.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion (or early-exit on V+/S+) — FLAGS field contains V+/S+ bits.

5.23 HW_TEST_EXT_TEMPERATURE_LOOPBACK (0x2E)

Test	Code 0x2E | Timeout 2500 ms | Sequences: none by default

Description	External temperature front-end loopback. Firmware drives the internal DACs across the full input range of the external NTC Wheatstone-bridge front-end and verifies the corresponding ADC reads track the stimulus. Covers all four EXT_TEMP channels.
Sequence(s)	Not in any default sequence — on B0/B1 only EXT_TEMP1/EXT_TEMP2 are populated, so the firmware DAC sweep would always saturate two channels. Available via SET_TEST for diagnostic use on boards with all four NTCs fitted.
Stimulus	None — internal DAC sweep is generated by firmware
Pass Criteria	V+ flag set (DAC→ADC tracking within tolerance for all four external temperature inputs). Long timeout (2500 ms) allows the DAC ramp + ADC settling sweep to complete.
ANLG[] Channels	ANLG[35]=EXT_TEMP1, ANLG[36]=EXT_TEMP2, ANLG[37]=EXT_TEMP3, ANLG[38]=EXT_TEMP4 (remapped from internal enum IDs 7..10).
Notes	Hardened by commit 06facf2 to protect NTC-resistance calculation against parameter/tolerance corner cases. A V- result here points to a broken bridge resistor, op-amp, or ADC mux on the external temperature path.

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.
2 →	RQST_SET_TEST (0x02)	value = 0x2E. m_reported_objects cleared. Test timeout = 2500 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Multiple EXT_TEMP analog reports during the DAC sweep.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion — FLAGS field contains V+/S+ bits.

5.24 HW_TEST_PUMP_AUTO (0x2F)

Test	Code 0x2F | Timeout 5000 ms | Sequences: B-Sample (B0/B1/B2)
Description	Automatic pump-driver verification. Firmware exercises every combination of the pump-driver control outputs (EN, LATCH, SEL) and sweeps the pump PWM duty cycle while monitoring the high-side switch currents and temperatures, plus the tachometer feedback.
Sequence(s)	B-Sample (B0/B1/B2) only — pump path not populated on A-sample
Stimulus	None during the sequence run. Optional manual control via RQST_SET_PUMP_DUTY (0x0A) and RQST_SET_PUMP_FREQ (0x0B) is available for interactive bring-up.
Pass Criteria	V+ flag set (every EN/LATCH/SEL combination produces expected HS-switch response across the duty sweep). ANLG[32]/ANLG[34] HS-switch temperatures within [-10, 85] °C; ANLG[39] PUMP_SPEED_TACH reports a non-zero count proportional to commanded duty.
ANLG[] Channels	ANLG[31]=PUMP_HS1_CURRENT, ANLG[32]=PUMP_HS1_TEMP, ANLG[33]=PUMP_HS2_CURRENT, ANLG[34]=PUMP_HS2_TEMP, ANLG[39]=PUMP_SPEED_TACH.
Notes	Pump must be connected to the board for the tachometer feedback to produce a non-zero count. With no pump attached, the firmware still validates the HS-switch outputs and a degraded-mode pass is reported by the latest hw_testing.c logic (commit 21dded5).

CAN Frame Exchange

Step	Frame	Content
1 →	RQST_SET_TEST_ENV (0x01)	value = 0x02 (REPORT_ENABLE). Sent once at sequence start.

2 →	RQST_SET_TEST (0x02)	value = 0x2F. m_reported_objects cleared. Test timeout = 5000 ms.
3 ←	RSPN_ANALOG (0x82) ×N	Pump HS-switch + tachometer reports during the EN/LATCH/SEL sweep.
4 ←	RSPN_TEST_STATUS (0x81)	After test completion — FLAGS field contains V+/S+ bits. Evaluated at end of timeout.

6. Oscilloscope Recommendations

For about a third of the automated tests, the firmware can only verify that internal configuration registers have been written correctly — it cannot observe the actual signals that come out of the gate-driver pins, the pump high-side switches or the internal DACs. For those tests an external oscilloscope provides the only direct view of correct operation.
This section identifies the tests where a scope is REQUIRED (no other means to verify the actual output), RECOMMENDED (useful to catch driver-side issues that wouldn't trigger a firmware failure), or OPTIONAL (purely digital protocols, scope adds little). It also proposes a fixed 4-channel probe layout that covers the whole B-sample campaign without moving probes between tests.

6.1 Recommended fixed 4-channel probe layout

All four signals listed below are 3.3 V CMOS / op-amp level at the MCU or driver-input pins on the B2 schematic — standard 10:1 passive probes are sufficient and no isolation is required. Use any nearby GND test point as the probe reference. Probe-point net names match the B2 schematic (sheet references in parentheses).

Channel	Signal (B2 net)	Probe location (schematic sheet)	Covers tests
CH1	DRV_UT_PWM	Phase U high-side PWM command, near MCU or driver-IC input (MCU.SchDoc / Driver.SchDoc)	0x18, 0x19, 0x1A, 0x1B
CH2	DRV_UB_PWM	Phase U low-side PWM command (MCU.SchDoc / Driver.SchDoc)	0x18, 0x19, 0x1A, 0x1B — deadtime visible vs CH1
CH3	PUMP_PWM	Pump driver PWM input (Ctrl.SchDoc / HighSideSwitch.SchDoc)	0x2F
CH4	DAC1	MCU DAC1 output pin (MCU.SchDoc)	0x25, 0x26, 0x2A, 0x2B, 0x2E, 0x28

If only a 2-channel scope is available, split the campaign into two passes: (1) CH1=DRV_UT_PWM + CH2=DRV_UB_PWM to cover the PWM frequency/duty/deadtime tests, then (2) CH1=PUMP_PWM + CH2=DAC1 to cover the pump and loopback tests.

6.2 Per-test scope guidance

Code	Test	Scope use	What to verify on the scope
0x10	ALL_DRIVERS_STATUS	recommended	DRV_xT_RDY / DRV_xB_RDY toggle as the firmware steps through power-on → enable → fault inject → clear.
0x14	ALL_DRIVERS_SETTING	optional	SPI traffic to drivers (SCK / MOSI / MISO) if SPI integrity is suspected.

0x18	ALL_DRIVERS_FREQUENCY	REQUIRED	DRV_UT_PWM frequency matches the swept setpoints — the firmware only confirms timer-register acceptance.
0x19	ALL_DRIVERS_DUTY	REQUIRED	DRV_UT_PWM duty cycle matches commanded value across the sweep.
0x1A	ALL_DRIVERS_DEADTIME	REQUIRED	Deadtime gap between DRV_UT_PWM falling edge and DRV_UB_PWM rising edge (not in default sequence — runs only on explicit request).
0x1B	ALL_POWER_MODULE	REQUIRED (HV)	Combined PWM + driver operation under HV bus (not in default sequence; bench safety required).
0x25	ENC_SINCOS_SIN_LOOPBACK	recommended	DAC ramp visible at the SIN+/SIN− differential input — confirms the analog front-end isn't clipping or DC-offset.
0x26	ENC_SINCOS_COS_LOOPBACK	recommended	Same checks applied to the COS leg.
0x28	DAC_OUT_AUTO	REQUIRED	DAC1 / DAC2 continuous waveform (sawtooth / triangular). Not in default sequence — invoked manually for DAC HW characterisation.
0x2A	POWER_UNIPLR_V_LOOPBACK	recommended	DAC1 sweep visible at the isolated input of the unipolar voltage-conditioning op-amp chain.
0x2B	POWER_BIPLR_V_LOOPBACK	recommended	DAC sweep covers positive and negative excursions of the bipolar conditioning chain (UV/WV phase-to-DCneg paths).
0x2D	POWER_MODULES_BY_PULSES	REQUIRED (HV)	Finite-pulse PWM burst (not in default sequence; bench safety required).
0x2E	EXT_TEMPERATURE_LOOPBACK	recommended	DAC ramp visible at the NTC Wheatstone bridge stimulus point —

			confirms the conditioning amplifier tracks the stimulus.
0x2F	PUMP_AUTO	REQUIRED	PUMP_PWM frequency/duty sweep, plus visual confirmation that PUMP_EN / PUMP_LATCH / PUMP_SEL combinations modulate PUMP_PWM_OUT correctly (probe HS-switch output if accessible).

Tests not listed (0x01–0x08, 0x0C, 0x1E–0x24, 0x29) are either purely DC, internal MCU readings or short digital-protocol exchanges where the scope adds little compared to the readings already captured in the .xlsx report.

6.3 Optional automated scope capture

code/scope_interface.py is a placeholder for SCPI control of a Rohde & Schwarz touch-panel scope (RTB / RTM / MXO families). It is inert by default. To enable automatic capture during a campaign:

Step	Action
1	Install dependencies: pip install pyvisa pyvisa-py
2	Identify the scope: front-panel model number (e.g. RTB2004, RTM3004, MXO54) and SCPI resource string (TCPIP0::<IP>::INSTR or USB0::0x0AAD::…).
3	Edit scope_interface.py: set SCOPE_MODEL, SCOPE_RESOURCE, then SCOPE_CAPTURE_ENABLED = True.
4	Adjust the per-test SCPI configuration in Scope.arm_for_test() and the measurement extraction in Scope.collect_after_test() to match the probe layout (§6.1).
5	Re-run run_tests.py — scope measurements / screenshot paths will be appended to each test result and end up in the ValidationReport_*.xlsx file.

Until the scope is wired in, the runner skips the hook silently and the campaign behaves exactly as before — only the operator-recorded scope screenshots (manual capture) need to be attached to the report.

6.4 Scope-assisted gate verification (ALL_DRIVERS_FREQUENCY / DUTY)

The firmware ALL_DRIVERS_FREQUENCY (0x18) and ALL_DRIVERS_DUTY (0x19) tests confirm only that the gate drivers report no fault (FLT clear) and are ready (RDY) while a frequency / duty sweep is applied to the six PWM channels. They do NOT measure the actual electrical characteristics of the isolated gate signals. The scope-assisted procedure below adds that electrical verification, switch by switch.

Tooling: code/run_gate_scope_check.py drives the firmware PWM test over CAN and controls a Rohde & Schwarz touch-scope (RTB2000 / RTM3000 / RTA4000 family) over USB (USBTMC via pyvisa + pyvisa-py — no vendor VISA install required). It prompts the operator to move a single voltage probe to each gate in turn, then captures and limit-checks the parameters.

Procedure — the tool iterates the six switches UTOP, UBOT, VTOP, VBOT, WTOP, WBOT. For each it (1) asks you to move the probe to that gate and press ENTER, (2) commands the firmware to drive the PWM (optionally at a fixed frequency or duty setpoint via RQST_SET_PWM_FREQ / RQST_SET_PWM_DUTY), (3) configures the scope vertical/timebase/trigger, (4) measures the seven parameters, and (5) records them. After all six it writes results/GateScopeReport_<SN>_<test>_<timestamp>.xlsx.

Parameters captured per switch and their default limits (+18 V / -3 V gate drive, 10:1 probe):

Parameter	Scope measurement	Default limit	Notes
Frequency	MEAS:MAIN FREQuency	setpoint ± 5 %	Checked vs commanded freq (freq test) or report-only (duty test)
Duty cycle	MEAS:MAIN PDCYcle	setpoint ± 5 %	Checked vs commanded duty (duty test) or report-only (freq test)
Rise time	MEAS:MAIN RTIMe	< 500 ns	10-90 % gate-voltage rise; tune to driver spec
Fall time	MEAS:MAIN FTIMe	< 500 ns	90-10 % gate-voltage fall
V-high	MEAS:MAIN HIGH	[16, 20] V	Positive gate level (+18 V nominal)
V-low	MEAS:MAIN LOW	[-5, -1] V	Negative gate level (-3 V nominal)
V-peak-to-peak	MEAS:MAIN PEAK	[18, 25] V	Total gate swing

Scope default setup: CH1, 10:1 probe attenuation, 4 V/div, +7.5 V offset (centres the +18/-3 V swing), DC coupling; timebase auto-set to ≈3 PWM periods from the frequency setpoint; edge trigger at +5 V. All overridable from the CLI (--scope-ch, --v-div, --offset, --probe-atten, --trig-level, --freq-khz, --duty, --resource).

Example: python run_gate_scope_check.py --test freq --freq-khz 16 --scope-ch 1 --unit-sn PCB-B1 (then move the probe and press ENTER for each of the six gates).

Scope model (confirmed): ROHDE & SCHWARZ RTM3004 (RTM3000 family). scope_rs.py is tuned for it — MEASurement1..7 are set up as parallel on-screen measurement slots (so the 7 parameters are also visible live on the scope during the manual probe moves), read back with MEASurement<m>:RESult:ACTual?, and the passive 10:1 probe attenuation is set via PROBe<m>:SETup:ATTenuation:MANual. full_setup() performs the complete pre-configuration (reset, vertical, timebase, edge trigger, measurement slots) before the first switch is measured. A --simulate mode returns canned readings for offline validation.

6.5 Scope-assisted power-module sweep (ALL_POWER_MODULE)

The firmware ALL_POWER_MODULE (0x1B) test energises all six power branches simultaneously while the operator sweeps three quantities manually. Unlike the gate-only checks of §6.4 (tests 0x18 / 0x19), which only confirm the drivers report no fault, this drives the real power stage and lets you characterise the gate signals across an operating grid, switch by switch.

Tooling: code/run_power_module_sweep.py drives HW_TEST_ALL_POWER_MODULE over CAN and the same Rohde & Schwarz touch-scope over USB. It pins the dead-band once (RQST SET_PWM_DT 0x07, value in ns), then for each switch walks every (frequency, duty) setpoint — pinning frequency via RQST_SET_PWM_FREQ (0x04, kHz) and duty via RQST_SET_PWM_DUTY (0x05) — and captures the same seven parameters as §6.4. Duty is entered as a percentage on the CLI but sent PER-UNIT (25 %→0.25, 50 %→0.5, 75 %→0.75) to match the firmware sweep range 0.0-1.0; the dead-band is entered and sent in nanoseconds (firmware sweep min 100 ns).

Default sweep — "10-10-30 kHz, 25-50-75 %, dead-band 1 µs": frequency 10/20/30 kHz (--pm-freq-min/step/max 10/10/30), duty 25/50/75 % (--pm-duty-min/step/max 25/25/75), dead-band 1000 ns (--deadband-ns 1000). That is a 3 × 3 = 9-setpoint grid per switch, 6 switches (54 points total). The duty points are deliberately kept away from 0 % / 100 % so every setpoint has real switching edges and is fully verdict-checked; widening the sweep to a 0 % or 100 % endpoint makes the tool report it as constant DC (frequency/duty become report-only, only the voltage levels are meaningful there). The report also appends Mean / Std(abs) / Std(rel) rows per measured column.

Per-setpoint verdict — the commanded frequency and duty are checked against the scope read-back (±5 %), and rise time / fall time / V-high / V-low / V-pp against the same default limits as §6.4. Scope pre-configuration, the edge-zoom second pass for accurate rise/fall, and the live on-screen MEAS1..7 slots are identical to the gate check.

⚠ Safety — 0x1B energises the real power branches. Ensure the DC-link / power stage is in the intended (de-energised or low-voltage) state for your bench before running. A --simulate mode returns canned readings for offline validation of the tool and report.

python run_power_module_sweep.py --scope-ch 1 --unit-sn PCB-B1 (default 10/20/30 kHz × 25/50/75 %, 1 µs dead-band; move the probe and press ENTER for each gate). Writes results/PowerModuleSweep_<SN>_<timestamp>.xlsx.

Integrated into the one-report verification — add --power-module to verify_pcba.py (see §7.4) and the per-setpoint records are folded into a "Power Module Sweep" sheet of the single ValidationReport workbook, alongside the automated campaign and the §6.4 gate checks.

7. Python Implementation

7.1 File Structure

File	Purpose
config.py	CAN interface parameters (interface, channel, bitrate), default sequence, unit S/N, operator name, timing constants.
hw_protocol.py	sfHwTestProtocol-compatible class: frame encoder/decoder, test dictionary, sequence lists, flag parsing. Mirrors STARK_FUTURE_hw_test_protocol.py.
pcan_interface.py	PCANBasic wrapper (ctypes-based, matching the original communication layer). Provides open/close, send_frame, recv_frame, flush_rx. Falls back to python-can if PCANBasic.dll is not found.
hw_test_runner.py	Sequence state machine engine (States 0–4). Implements the same logic as _ThreadSequenceExecute() in the original. Runs synchronously for automated non-interactive testing. Accumulates m_reported_objects, evaluates V+/S+, checks analog limits.
hw_test_criteria.py	ANLG channel name map and numeric limits (5V_RAIL, temperatures, phase currents). Used by hw_test_runner for secondary pass/fail checking.
run_tests.py	Entry point CLI. Parses --sequence, --unit-sn, --operator. Calls execute_campaign(), prints summary, triggers report generation.
generate_report.py	Generates a formatted Excel workbook (.xlsx) from the results list using openpyxl. Sheets: Summary, Test Results, Analog Readings, Conclusions (with sign-off block).
requirements.txt	Python package dependencies: python-can>=4.0.0, openpyxl>=3.1.0.

7.2 Communication Layer

The original code uses the PCANBasic.dll Windows driver directly through a ctypes wrapper (PCANBasic.py from PEAK). The new automated code adapts this to maintain the same communication behaviour:

- pcan_interface.py attempts to import PCANBasic. If the DLL is present, it uses TPCANMsg with PCAN_MESSAGE_STANDARD (11-bit frames), PCAN_USBBUS1, and PCAN_BAUD_250K — identical to the original.
- If PCANBasic.dll is not available, the module falls back to python-can with interface='pcan', channel='PCAN_USBBUS1', bitrate=250000, which targets the same hardware through a different API.
- Frame structure is identical in both paths: 8-byte standard (non-FD) frames on arbitration ID 0x100.

7.3 State Machine vs. Original Threading

The original tool used three threads (listening, sequence, macro) running simultaneously to support interactive use. The automated runner simplifies this to a synchronous loop since no interactive user input is needed during an automated campaign. The state transitions (States 0-4) and timing (20 ms quantum) are preserved exactly.

7.4 How to Run

Prerequisites: PEAK PCAN-USB driver + board on HW-test firmware. For the optional scope phase: pip install pyvisa pyvisa-py pyusb (pure-Python USBTMC; or install R&S-VISA / NI-VISA and use --backend @ivi).

PRIMARY WORKFLOW — verify_pcba.py runs the complete PCBA verification and writes ONE report file (results/ValidationReport_<SN>_<timestamp>.xlsx) containing every result: automated campaign + manual gate scope checks.

python verify_pcba.py --unit-sn PCB-B1 # automated campaign only

python verify_pcba.py --unit-sn PCB-B1 # Phase 1 only (self tests, ~17, unattended)

python verify_pcba.py --unit-sn PCB-B1 --power-module # Phase 1 + Phase 2 (power modules scope sweep)

python verify_pcba.py --unit-sn PCB-B1 --loopback # Phase 1 + Phase 3 (loopback DAC, interactive)

python verify_pcba.py --unit-sn PCB-B1 --all # Phase 1 + Phase 2 + Phase 3 (full PCBA campaign)

python verify_pcba.py --unit-sn PCB-B1 --scope --freq-khz 16 --duty 50 # + manual gate scope phase

python verify_pcba.py --unit-sn PCB-B1 --scope --power-module # + manual gate scope AND power-module sweep phases

python verify_pcba.py --unit-sn PCB-B1 --no-prompts # skip all manual prompts

Before the first scope run, validate the scope link with the self-test (independent of CAN / firmware / probe):

python scope_check.py # auto-discover, single read

python scope_check.py --watch # live continuous read while probing

python scope_check.py --list-only # just enumerate VISA resources

python scope_check.py --resource "TCPIP::192.168.1.50::INSTR" # LAN if USB won't enumerate

Scope connection note (RTM3004 on Windows): pyvisa-py USBTMC usually will NOT enumerate the scope over USB unless a libusb driver is bound to it (Zadig) — and that conflicts with the R&S USB driver. Two reliable options: (a) RECOMMENDED for USB — install R&S VISA (RsVisaSetup, free from rohde-schwarz.com) and run with --backend @ivi; or (b) use LAN — set the RTM3004 to a network address (Setup → Interfaces → LAN), then pass --resource "TCPIP::<scope-ip>::INSTR" (works out-of-the-box with pyvisa-py, no driver). Verify with scope_check.py first.

Legacy / focused entry points (still available):

python run_tests.py --sequence B2 --unit-sn PCB-B1 # automated campaign only (own report)

python run_gate_scope_check.py --test freq --freq-khz 16 # standalone scope-only gate check

python run_power_module_sweep.py --scope-ch 1 # standalone power-module sweep (0x1B), own report

Outputs: results/test_results_<SN>_<timestamp>.csv and the single results/ValidationReport_<SN>_<timestamp>.xlsx (sheets: Summary, Test Results, Analog Readings, Gate Scope Checks, Power Module Sweep, Conclusions).